```
from google.colab import files
uploaded = files.upload()
```

```
<IPython.core.display.HTML object>
```

```
Saving Mall_Customers - Mall_Customers.xlsx.csv to Mall_Customers -
Mall_Customers.xlsx.csv
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

df = pd.read_csv('Mall_Customers - Mall_Customers.xlsx.csv')
df.info()
print(df.head())

sns.pairplot(df)

features = df.iloc[:, [3, 4]].values

from sklearn.cluster import KMeans
model = KMeans(n_clusters=5, random_state=42, n_init=10)
model.fit(features)
print(model)

Final = df.iloc[:, [3, 4]].copy()
Final['label'] = model.predict(features)
print(Final.head())

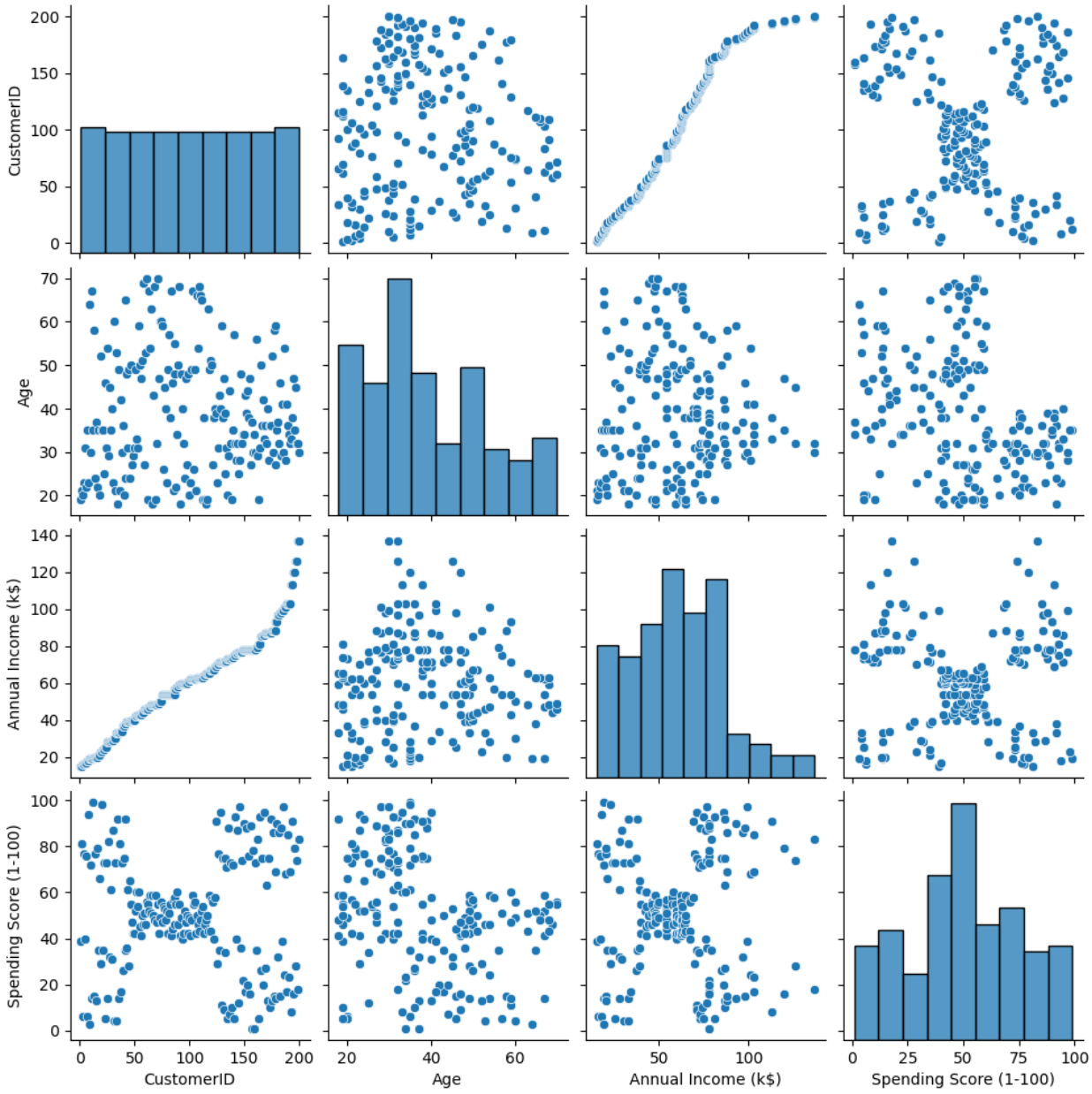
sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
    .map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)")
\
    .add_legend()
plt.show()

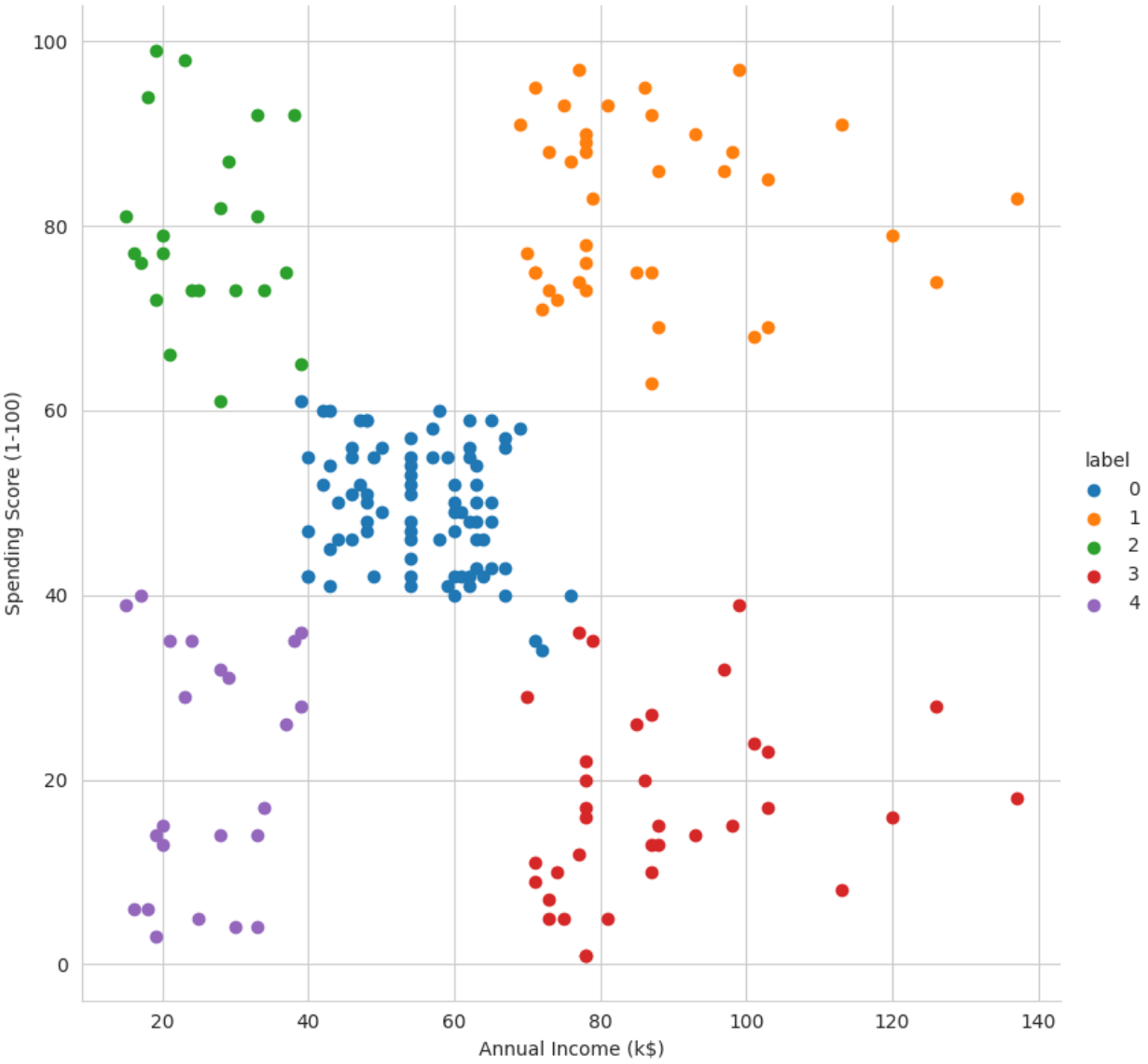
features_el = df.iloc[:, [2, 3, 4]].values
wcss = []

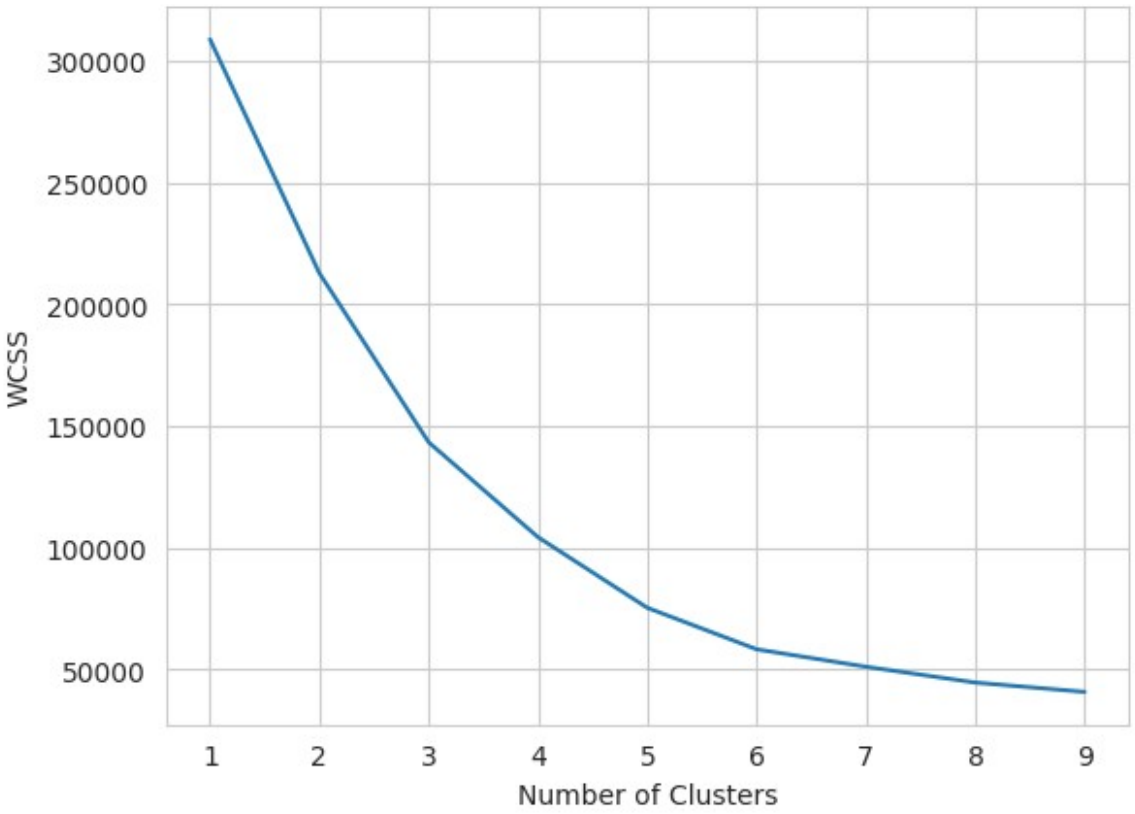
for i in range(1, 10):
    model = KMeans(n_clusters=i, random_state=42, n_init=10)
    model.fit(features_el)
    wcss.append(model.inertia_)

plt.plot(range(1, 10), wcss)
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   CustomerID              200 non-null    int64  
 1   Gender                  200 non-null    object 
 2   Age                     200 non-null    int64  
 3   Annual Income (k$)      200 non-null    int64  
 4   Spending Score (1-100)  200 non-null    int64  
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
KMeans(n_clusters=5, n_init=10, random_state=42)
   Annual Income (k$)  Spending Score (1-100)  label
0                  15                      39      4
1                  15                      81      2
2                  16                       6      4
3                  16                      77      2
4                  17                      40      4
```


CustomerID
Age
Annual Income (k$)
Spending Score (1-100)
CustomerID
Age
Annual Income (k$)
Spending Score (1-100)


Spending Score (1-100)
Annual Income (k$)
label
0
1
2
3
4
0
20
40
60
80
100
20
40
60
80
100
120
140


300000
250000
200000
150000
100000
50000
WCSS
1
2
3
4
5
6
7
8
9
Number of Clusters